

Luis Alejandro Rodríguez Arenas

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EDUCATION

Universidad de los Andes, Bogotá, Colombia B.S. Mechanical Engineering, Minor in Computational Mathematics. GPA: 4.4 / 5.0	Ongoing
Universidad Nacional de Colombia, Bogotá, Colombia Undergraduate Coursework in Mechatronics Engineering. GPA: 4.6 / 5.0	2023
Colegio San Carlos, Bogotá, Colombia High School Diploma. SAT: 1320 (660 Math, 660 and Writing). ICFES Saber 11: Top 1% National Score	2017 – 2023

EXPERIENCE

Course Monitor — Mechanical Systems Dynamics, Universidad de los Andes Assisted instruction and grading for an undergraduate course in rigid-body dynamics and engineering mechanics.	2026
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PROJECTS

Ongoing CCCD2026 Rocketry Program — Director, SpaceTech Rocketry Leading two multidisciplinary teams developing fiberglass and carbon-fiber rocket platforms while overseeing avionics, telemetry downlink, manufacturing, and flight-test operations.	
Ongoing Experimental Rocketry Development Program — Director, SpaceTech Rocketry Leading SpaceTech's transition toward experimental rocketry through solid-propellant research, safety framework development, and long-term preparation for the Latin American Space Challenge.	
2026 CanSat World Championship 2026 — Fuselage Development Director Led development of a fiberglass airframe that reduced projected enclosure mass by 77% relative to an equivalent aluminum design while supporting avionics, payload, and recovery-system integration.	
2025 CCCD2025 Rocketry Program — Aerodynamics & Mechanical Design Director Led aerodynamics and mechanical design for a fiberglass sounding rocket, validating performance through CFD, structural simulation, materials testing, and subsystem integration activities.	
2024 Ion Thruster Research Project Developed analytical models of ion acceleration and charge transport while supporting feasibility studies for low-power electric propulsion systems.	

AWARDS

Act In Space 2026 Won Act In Space Bogotá and advanced to the international finals in Bordeaux, France.	2026
Banco de la República Academic Excellence Scholarship Awarded for achieving one of the highest national scores in the ICFES Saber 11 examination.	2024
Universidad Nacional Academic Excellence Scholarship Awarded in recognition of outstanding academic performance among graduating high-school students.	2023
International Olympiad on Astronomy and Astrophysics 2022 Represented Colombia at the international competition held in Kutaisi, Georgia.	2022

SKILLS

Engineering Software	SolidWorks, Inventor, Fusion 360, OnShape, ANSYS
Programming	Python, C++, MATLAB, Git, Linux
Embedded	Arduino, ESP32, Raspberry Pi, Telemetry Systems
Manufacturing	Carbon Fiber, Fiberglass, Composite Layups, 3D Printing
Aerospace	Rocketry, Flight Dynamics, Avionics, Electric Propulsion
Languages	Spanish (Native), English (C1), French (B2)